

| Programiz                    | Part |
|------------------------------|------|
| DSA Intro                    | 1    |
| Getting Started with DSA     |      |
| What is an algorithm         |      |
| Data Structure and Types     |      |
| Why learn DSA?               |      |
| Asymptotic Notations         |      |
| Master Theorem               | 2    |
| Divide and Conquer Algorithm |      |
| Data Structures-I            |      |
| Stack                        |      |
| Queue                        |      |
| Types of Queue               |      |
| Circular Queue               |      |
| Priority Queue               |      |
| Deque                        |      |
| Data Structures-2            |      |
| Linked List                  |      |
| Linked List Operations       |      |
| Types of Linked List         |      |
| Hash Table                   |      |
| Heap Data Structure          |      |
| Fibonacci Heap               |      |
| Decrease Key                 | 3    |
| Delete Node                  | 3.1  |
| Tree based DSA               |      |
| Tree Data Structure          |      |
| Tree Traversal               |      |
| Binary Tree                  |      |
| Full Binary Tree             |      |
| Perfect Binary Tree          |      |
| Complete Binary Tree         |      |
| Balanced Binary Tree         |      |
| AVL Tree                     |      |
| Tree based DSA               |      |
| B Tree                       |      |
| Insertion in a B Tree        |      |
| Deletion from a B Tree       |      |

|                                  |     |
|----------------------------------|-----|
| B+ Tree                          |     |
| Insertion on a B+ Tree           |     |
| Deletion from a B+ Tree          |     |
| Red-Black Tree                   |     |
| Red-Black Tree Insertion         |     |
| Deletion from a Red Black Tree   | 3.2 |
| Graph Based DSA                  |     |
| Graph Data Structure             |     |
| Spanning Tree                    |     |
| Strongly Connected Components    |     |
| Adjacency Matrix                 |     |
| DFS Algorithm                    |     |
| Breadth first Search             |     |
| Bellam Ford Algo                 |     |
| Sorting and Searching Algorithms |     |
| Bubble Sort                      |     |
| Selection Sort                   |     |
| Insertion Sort                   | 3.3 |
| Merge Sort                       |     |
| Quick Sort                       |     |
| Coutning Sort                    |     |
| Radix Sort                       |     |
| Buckect Sort                     |     |
| Heap Sort                        |     |
| Shell                            |     |
| Linear Search                    |     |
| Binary Search                    | 3.4 |
| Greedy Algorithm                 |     |
| Greedy Algorithm                 |     |
| Ford Fulkerson Algorithm         |     |
| Dijkstra Algorithm               |     |
| Prim's Algorithn                 |     |
| Huffman Coding                   |     |
| Dynamic Programming              |     |
| Dynamic Programming              |     |
| Floyd Warshall Algorithm         |     |
| Longest Common Sequence          |     |
|                                  | 4   |
| Other Algorithm                  |     |
| back tracking Algorithm          |     |

## Rabin Karp Algorithm

## Element of Programming Interview

### # Topics

The Interview

Getting Ready

Strategies for a great interview

Conducting an interview

Problem solving

Problems

Primitive Types 11

Arrays 19

Strings 13

LinkedList 13

Stack and queues 10

Binary Trees 17

Heaps 7

Searching 10

Hash Table 14

Sort 10

BST 13

Recursion 11

DP 12

Greedy Algorithm 8

Graphs 9

Parallel Computing 9

Domain Specific

Design Pattern 19

1. Design a spell checker

2. Design a solution to the stemming Problem

3. Plagiarism detector

4. Pair users by attribute

5. Design a system for detecting copyright infringement

6. Design TEX

7. Design a search engine

8. Implement Page Rank

9. Design TeraSort and PetaSort

10. Implement distributed throttling

11. Design a scalable priority system

12. Create Photomosaics

13. Implement Mileage Run

14. Implement Connexus

15. Design an online advertising system

16. Design a recommendation system
17. Design A distributed file system
18. Design the world wide web
19. Estimate hardware cost of photo shaing app

Language Questions (C++) 11

References and pointers

Pass by reference vs pass by value

Smart pointers

Iterators

Constructors

Default methods

malloc(), free(), new, delete

Strings,

Push\_back and implace\_back()

Fast function calls

Template functions

Run-time type identification

Object Oriented Fdesign 7

Template Methods vs Strategy

Observer Pattern

Push vs Pull observer pattern

Singletons and Flyweights

Adapters

Creational Patterns

Libraries and design pattern

Common Tools 12

Merging in a version control system

Hooks

Is scripting is more efficient

Polymorphism with scripting language

Dependency analysis

ANT vs Maven

SQL vs No SQL

Normalization

SQL design

IP, TCP, HTTP

DNS

Honors Class 40

Compute the GCD

Find missing positive entry

Buy and sell a stock k times  
Compute the maximum product of all the entires but one  
compute the longest contiguous increasing subarray  
Rotate an array  
Identity position attacked by rook  
Justify text  
Implement list zipping  
Copy a posting list  
Compute the longest substring with matching parens  
Compute the maximum sliding windos  
Implement a postorder traversal without recursion  
Compute fair bonuses  
Search a sorted array of unknown length  
search in two sorted array  
find the kth largest element - large n, small k  
Find an element that appears only once  
Find the line through the most points  
Find the shortest unique prefic  
Find the most visited pages in a windos  
Convert a sorted doubly linked list into BST  
Convert a BST to a sorted double link list  
Merge two BST  
The view from above  
Implement regular experssion matching  
Synthesize an epression  
Count inversions  
Draw the skyline  
Measure with defective jugs  
Compute the maximum subarray sum in a circular array  
Determine the critical height  
Find the maximum 2D subarray  
Implement huffman coding  
Trapping water  
Search for a pair-sum in an abs-sorted array  
The heavy hitter problem  
Find the lingest subarray whose sum  $\leq k$   
Road network  
Test if arbitrage is possible